

CIL MT - Paper-I

Questions and answers only. No explanations.

117 questions across 13 chapters

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Lexicon, Grammar, and Sentence Correction

English - 7 questions

1. Choose the word that is most nearly SYNONYMOUS in meaning to the word in capital letters:

EPHEMERAL

- (A) Eternal
- (B) Transient
- (C) Abundant
- (D) Ethereal

Answer: (B) Transient

2. Choose the word that is most nearly ANTONYMOUS in meaning to the word in capital letters:

CONSPICUOUS

- (A) Obvious
- (B) Unobtrusive
- (C) Prominent
- (D) Ostentatious

Answer: (B) Unobtrusive

3. In the following sentence, identify the segment that contains a grammatical error. If there is no error, choose 'No error':

'Neither the supervisor nor the miners was aware of the sudden buildup of methane gas in the gallery.'

- (A) Neither the supervisor nor
- (B) the miners was aware of
- (C) the sudden buildup of methane gas
- (D) No error

Answer: (B) the miners was aware of

4. Identify the option that represents the grammatically correct and most appropriate rewrite of the underlined section:

'The CEO along with his board members, <u>are visiting the coal washery facility tomorrow morning</u>.'

- (A) are visiting the coal washery facility tomorrow morning
- (B) is visiting the coal washery facility tomorrow morning
- (C) have visited the coal washery facility tomorrow morning
- (D) were visiting the coal washery facility tomorrow morning

Answer: (B) is visiting the coal washery facility tomorrow morning

5. Fill in the blanks with the correct pair of prepositions:

'The mining equipment must comply _____ the latest safety standards, and operations must be carried _____ in accordance with the regulatory act.'

- (A) with, out
- (B) by, on
- (C) to, off
- (D) with, through

Answer: (A) with, out

6. What is the meaning of the idiomatic expression underlined in the sentence below?

'The project manager was asked to <u>cut corners</u> to finish the dispatch reports before the auditing committee arrived.'

- (A) To work overtime and put extra effort
- (B) To perform a task in a sloppy, cheap, or fast way by ignoring rules or standards
- (C) To divide a square piece of land into equal parts
- (D) To resolve a complex dispute between two labor unions

Answer: (B) To perform a task in a sloppy, cheap, or fast way by ignoring rules or standards

7. Select the correct word to fill in the blank to maintain grammatical correctness and sentence coherence:
'The persistent decline in international coal prices has had a severe _____ on the company's export profit margins.'

(A) affect
(B) effect
(C) effective
(D) affective

Answer: (B) effect

Reading Comprehension and Text Structuring

English - 5 questions

8. Read the following passage carefully and answer the question that follows:

'The energy transition is not a binary switch but a multi-decade journey of technological, infrastructural, and economic adaptation. For developing nations like India, coal remains the bedrock of baseload electricity generation, shielding the grid from the intermittency of solar and wind inputs. While renewable capacity has witnessed exponential growth, industrial operations like steel manufacturing and cement production cannot be electrified with current battery technologies. This reality has catalyzed investments in clean-coal technologies (CCT) such as supercritical boilers and coal gasification, allowing nations to lower their carbon intensity while securing their industrial lifelines. Thus, premature abandonment of fossil fuels without robust backup structures risks grid instability and economic deceleration.'

According to the passage, why can't coal be completely and immediately replaced by renewable energy sources in developing nations like India?

(A) Developing nations lack the financial capital to build any wind or solar farms.
(B) Coal provides reliable baseload power, and heavy industries like steel and cement cannot be electrified with current battery storage technology.
(C) Coal has a lower carbon intensity than wind and solar power systems.
(D) International climate treaties forbid developing nations from adopting green energy technologies.

Answer: (B) Coal provides reliable baseload power, and heavy industries like steel and cement cannot be electrified with current battery storage technology.

9. Based on the passage provided in Question 1, what is the role and objective of clean-coal technologies (CCT)?

(A) To immediately eliminate all coal-fired power stations globally.
(B) To allow countries to completely bypass the transition to renewable energy sources.
(C) To lower carbon intensity and emissions of coal usage while securing energy security for industrial operations.
(D) To manufacture advanced high-capacity lithium-ion batteries for heavy industries.

Answer: (C) To lower carbon intensity and emissions of coal usage while securing energy security for industrial operations.

10. Rearrange the following sentences in a logical sequence to form a coherent paragraph:

P. Consequently, grid operators must deploy fast-ramping backup sources to stabilize the voltage fluctuations.

Q. Unlike traditional thermal plants, solar panels and wind turbines produce electricity only when weather conditions are favorable.

R. The integration of high-penetration renewable energy presents a massive challenge for electrical grids.

S. This inherent intermittency can cause sudden dips in power supply on cloudy or windless days.

Choose the correct sequence from the options below:

(A) RQSP
(B) RSPQ
(C) QSRP
(D) PQSR

Answer: (A) RQSP

11. In the following question, some parts of the sentence have been jumbled up. Rearrange these parts (P, Q, R, S) to form a grammatically correct and meaningful sentence:

He decided to undergo a rigorous training program (P) / in order to enhance his operational efficiency (Q) / before taking charge of the deep shaft underground mine (R) / as the newly appointed manager (S).

- (A) PQRS
- (B) PSRQ
- (C) RPSQ
- (D) QPRS

Answer: (A) PQRS

12. Choose the correct words to complete the sentence, ensuring grammatical correctness and appropriate meaning:

'The labor union leaders were _____ by the management's uncompromising stance, and their subsequent refusal to negotiate _____ the ongoing strike.'

- (A) exasperated, prolonged
- (B) appeased, resolved
- (C) pacified, ignited
- (D) elated, terminated

Answer: (A) exasperated, prolonged

General Science, Technology, and Computer Awareness

General Awareness - 7 questions

13. Which of the following classes of coal has the highest carbon content, the highest calorific value, and the lowest moisture content?

- (A) Lignite
- (B) Peat
- (C) Bituminous
- (D) Anthracite

Answer: (D) Anthracite

14. In computer networking, which of the following application layer protocols is used to dynamically assign IP addresses to devices on a network?

- (A) DNS (Domain Name System)
- (B) DHCP (Dynamic Host Configuration Protocol)
- (C) SMTP (Simple Mail Transfer Protocol)
- (D) ARP (Address Resolution Protocol)

Answer: (B) DHCP (Dynamic Host Configuration Protocol)

15. The 'PARAM Ganga' supercomputer, established under India's National Supercomputing Mission (NSM), is installed at which of the following prestigious institutes?

- (A) IISc Bangalore
- (B) IIT Bombay
- (C) IIT Roorkee
- (D) IIT Kharagpur

Answer: (C) IIT Roorkee

16. What is the primary physical phenomenon utilized in the functioning of Optical Fiber Cables (OFC) used in high-speed internet communications?

- (A) Refraction of Light
- (B) Total Internal Reflection
- (C) Diffraction of Light
- (D) Interference of Light

Answer: (B) Total Internal Reflection

17. Who is credited with the invention of the World Wide Web (WWW) in 1989, which revolutionized information sharing across the globe?

- (A) Vint Cerf
- (B) Tim Berners-Lee
- (C) Steve Jobs
- (D) Alan Turing

Answer: (B) Tim Berners-Lee

18. Which of the following security protocols is universally used to establish an encrypted and secure link between a web server and a browser, denoted by 'https' in the URL?

- (A) FTP (File Transfer Protocol)
- (B) SSH (Secure Shell)
- (C) SSL/TLS (Secure Sockets Layer / Transport Layer Security)
- (D) WEP (Wired Equivalent Privacy)

Answer: (C) SSL/TLS (Secure Sockets Layer / Transport Layer Security)

19. The Nobel Prize in Physics 2023 was awarded to Pierre Agostini, Ferenc Krausz, and Anne L'Huillier for their pioneering work in which scientific domain?

- (A) Quantum Entanglement and Information Science
- (B) Experimental methods that generate attosecond pulses of light for the study of electron dynamics
- (C) Pioneering contributions to our understanding of complex physical systems and climate modeling
- (D) Discovery of the Higgs Boson particle

Answer: (B) Experimental methods that generate attosecond pulses of light for the study of electron dynamics

Polity, History, Geography, and Indian Economy

General Awareness - 7 questions

20. Under the Indian Constitution, the 'Right to Property' was deleted from the list of Fundamental Rights and made a simple legal right under Article 300A by which Constitutional Amendment Act?

- (A) 42nd Amendment Act, 1976
- (B) 44th Amendment Act, 1978
- (C) 52nd Amendment Act, 1985
- (D) 86th Amendment Act, 2002

Answer: (B) 44th Amendment Act, 1978

21. Which of the following geographical regions contains the oldest and most significant coalfields of India, namely the Raniganj and Jharia coalfields?

- (A) Narmada River Valley
- (B) Damodar River Valley
- (C) Mahanadi River Valley
- (D) Godavari River Valley

Answer: (B) Damodar River Valley

22. Who presided over the historic Lahore Session of the Indian National Congress in December 1929, where the resolution of 'Purna Swaraj' (Complete Independence) was adopted?

- (A) Mahatma Gandhi
- (B) Subhas Chandra Bose
- (C) Jawaharlal Nehru
- (D) Motilal Nehru

Answer: (C) Jawaharlal Nehru

23. Which of the following bodies is responsible for officially presenting the 'Union Budget' of India, and under which Constitutional Article is it presented as the 'Annual Financial Statement'?

- (A) NITI Aayog, Article 110
- (B) Ministry of Finance, Article 112
- (C) Reserve Bank of India, Article 280
- (D) Finance Commission, Article 360

Answer: (B) Ministry of Finance, Article 112

24. In the Indian economy, which of the following is designated as the 'base year' for the calculation of the Consumer Price Index (CPI) and Index of Industrial Production (IIP)?

- (A) 2004-05
- (B) 2011-12
- (C) 2015-16
- (D) 2018-19

Answer: (B) 2011-12

25. Which major sea port of India, located on the East Coast in Odisha, is highly crucial for the massive export of iron ore and coal to East Asian countries like Japan?
- (A) Haldia Port
 - (B) Visakhapatnam Port
 - (C) Paradip Port
 - (D) Ennore Port
- Answer: (C) Paradip Port**
26. The famous Ellora Caves, a UNESCO World Heritage site consisting of 34 major rock-cut cave temples representing Hinduism, Buddhism, and Jainism, were constructed primarily under the patronage of which dynasty?
- (A) Chola Dynasty
 - (B) Gupta Dynasty
 - (C) Rashtrakuta Dynasty
 - (D) Pallava Dynasty
- Answer: (C) Rashtrakuta Dynasty**

Coal Sector in India, Climate Change, and SDGs

General Awareness - 7 questions

27. Which of the following Indian states has the highest estimated geological reserves of coal as per the latest inventory of the Geological Survey of India (GSI)?
- (A) Chhattisgarh
 - (B) Odisha
 - (C) Jharkhand
 - (D) Madhya Pradesh
- Answer: (C) Jharkhand**
28. Which subsidiary of Coal India Limited (CIL) holds the prestigious title of being the single largest coal-producing company in India, operating major projects like Gevra and Dipka?
- (A) Mahanadi Coalfields Limited (MCL)
 - (B) South Eastern Coalfields Limited (SECL)
 - (C) Northern Coalfields Limited (NCL)
 - (D) Eastern Coalfields Limited (ECL)
- Answer: (B) South Eastern Coalfields Limited (SECL)**
29. Under India's Nationally Determined Contributions (NDCs) updated in 2022, what target year has India officially committed to for achieving 'Net-Zero' greenhouse gas emissions?
- (A) 2050
 - (B) 2060
 - (C) 2070
 - (D) 2100
- Answer: (C) 2070**
30. Which United Nations Sustainable Development Goal (SDG) explicitly aims to ensure 'Affordable and Clean Energy' for all by the year 2030?
- (A) SDG 6
 - (B) SDG 7
 - (C) SDG 9
 - (D) SDG 13
- Answer: (B) SDG 7**
31. In the context of clean coal technologies, the Ministry of Coal has launched a National Mission to achieve coal gasification of how many million tonnes (MT) of coal by the year 2030?
- (A) 50 MT
 - (B) 100 MT
 - (C) 150 MT
 - (D) 200 MT
- Answer: (B) 100 MT**

- 32.** The 'Kyoto Protocol', an international treaty adopted in 1997 under the UNFCCC, is famously known for introducing which of the following market-based compliance mechanisms?
- (A) Loss and Damage Fund
 - (B) Clean Development Mechanism (CDM)
 - (C) Global Green Growth Institute
 - (D) Technology Executive Committee
- Answer: (B) Clean Development Mechanism (CDM)**
- 33.** Which of the following coal mining techniques involves leaving solid pillars of coal to support the roof of the mine, while excavating a network of intersecting galleries, and is widely practiced in India's underground coal operations?
- (A) Longwall Mining
 - (B) Board and Pillar (Room and Pillar) Mining
 - (C) Opencast Strip Mining
 - (D) Hydraulic Mining
- Answer: (B) Board and Pillar (Room and Pillar) Mining**

Current Affairs, Awards, Sports, and National-International Organizations

General Awareness - 7 questions

- 34.** Which of the following thermal power plants is currently the largest operational coal-fired super thermal power station in India, with an installed capacity of 4,760 MW, owned and operated by NTPC?
- (A) Mundra Ultra Mega Power Project
 - (B) Vindhyachal Super Thermal Power Station
 - (C) Sasan Ultra Mega Power Project
 - (D) Tiroda Thermal Power Station
- Answer: (B) Vindhyachal Super Thermal Power Station**
- 35.** The 'Kuno National Park', which recently made global headlines for the historic transcontinental reintroduction of African Cheetahs to India, is located in which state?
- (A) Rajasthan
 - (B) Madhya Pradesh
 - (C) Gujarat
 - (D) Maharashtra
- Answer: (B) Madhya Pradesh**
- 36.** The G20 (Group of Twenty) officially admitted which of the following multilateral organizations as its permanent 21st member during the New Delhi Summit in September 2023 under India's Presidency?
- (A) Association of Southeast Asian Nations (ASEAN)
 - (B) African Union (AU)
 - (C) South Asian Association for Regional Cooperation (SAARC)
 - (D) European Free Trade Association (EFTA)
- Answer: (B) African Union (AU)**
- 37.** The 'PM-KUSUM' scheme, launched by the Ministry of New and Renewable Energy (MNRE), primarily aims to achieve which of the following objectives?
- (A) Financial security for small and marginal traders
 - (B) Solarization of agricultural water pumps and installation of grid-connected solar plants
 - (C) Providing LPG connections to rural households below poverty line
 - (D) Digital literacy training for rural self-help groups
- Answer: (B) Solarization of agricultural water pumps and installation of grid-connected solar plants**
- 38.** Who was awarded India's highest literary honor, the 57th Jnanpith Award, in 2022 for outstanding contribution to Konkani literature?
- (A) Damodar Mauzo
 - (B) Nilmani Phookan
 - (C) Amitav Ghosh
 - (D) Akkitham Achuthan Namboothiri
- Answer: (A) Damodar Mauzo**

39. Which nation hosted the 2024 ICC Men's T20 World Cup jointly with the West Indies, marking the first time an ICC major tournament was held in this country?
- (A) Canada
 - (B) United States of America
 - (C) England
 - (D) South Africa

Answer: (B) United States of America

40. The classic national historical masterwork 'Anandamath', which contains the lyrics of India's national song 'Vande Mataram', was written by which iconic author?
- (A) Rabindranath Tagore
 - (B) Bankim Chandra Chattopadhyay
 - (C) Sarojini Naidu
 - (D) Subhash Chandra Bose

Answer: (B) Bankim Chandra Chattopadhyay

CIL MT Mock Sheet-1 (General Aptitude)

Mock Tests - 32 questions

41. A can complete a piece of work in 12 days, and B can complete the same work in 15 days. If they work together, how many days will they take to finish the work?
- (A) 6 days
 - (B) 6.67 days
 - (C) 7 days
 - (D) 5.5 days

Answer: (B) 6.67 days

42. Pointing to a photograph, a man said, 'I have no brother or sister but that man's father is my father's son.' Who is in the photograph?
- (A) His father
 - (B) His nephew
 - (C) His son
 - (D) Himself

Answer: (C) His son

43. Choose the word that is most nearly SYNONYMOUS with 'PRUDENT'.
- (A) Reckless
 - (B) Wise
 - (C) Impassive
 - (D) Frugal

Answer: (B) Wise

44. Which Indian PSU is the single largest coal producer in the world?
- (A) NTPC Limited
 - (B) Coal India Limited (CIL)
 - (C) ONGC
 - (D) SAIL

Answer: (B) Coal India Limited (CIL)

45. A sum of money doubles itself in 5 years at simple interest. In how many years will it become fourfold?
- (A) 10 years
 - (B) 12 years
 - (C) 15 years
 - (D) 20 years

Answer: (C) 15 years

46. Fill in the blank: The management trainee candidates _____ completed their submission on time.
- (A) has
 - (B) have
 - (C) is
 - (D) having

Answer: (B) have

47. Which year was Coal India Limited founded as a state-owned enterprise?

- (A) 1972
- (B) 1975
- (C) 1980
- (D) 1991

Answer: (B) 1975

48. What is the value of x if the average of 5 consecutive odd numbers is 38?

- (A) 34
- (B) 38
- (C) 40
- (D) 42

Answer: (B) 38

49. In a certain code, 'COAL' is written as 'DQDP'. How is 'GOLD' written in that same code?

- (A) HPME
- (B) HQOF
- (C) HPMF
- (D) IPNF

Answer: (A) HPME

50. What is the value of x if the average of 5 consecutive odd numbers is 42?

- (A) 38
- (B) 42
- (C) 44
- (D) 46

Answer: (B) 42

51. What is the value of x if the average of 5 consecutive odd numbers is 46?

- (A) 42
- (B) 46
- (C) 48
- (D) 50

Answer: (B) 46

52. What is the value of x if the average of 5 consecutive odd numbers is 50?

- (A) 46
- (B) 50
- (C) 52
- (D) 54

Answer: (B) 50

53. What is the value of x if the average of 5 consecutive odd numbers is 54?

- (A) 50
- (B) 54
- (C) 56
- (D) 58

Answer: (B) 54

54. What is the value of x if the average of 5 consecutive odd numbers is 58?

- (A) 54
- (B) 58
- (C) 60
- (D) 62

Answer: (B) 58

55. What is the value of x if the average of 5 consecutive odd numbers is 62?

- (A) 58
- (B) 62
- (C) 64
- (D) 66

Answer: (B) 62

56. What is the value of x if the average of 5 consecutive odd numbers is 66?

- (A) 62
- (B) 66
- (C) 68
- (D) 70

Answer: (B) 66

57. What is the value of x if the average of 5 consecutive odd numbers is 70?

- (A) 66
- (B) 70
- (C) 72
- (D) 74

Answer: (B) 70

58. What is the value of x if the average of 5 consecutive odd numbers is 74?

- (A) 70
- (B) 74
- (C) 76
- (D) 78

Answer: (B) 74

59. What is the value of x if the average of 5 consecutive odd numbers is 78?

- (A) 74
- (B) 78
- (C) 80
- (D) 82

Answer: (B) 78

60. What is the value of x if the average of 5 consecutive odd numbers is 82?

- (A) 78
- (B) 82
- (C) 84
- (D) 86

Answer: (B) 82

61. What is the value of x if the average of 5 consecutive odd numbers is 86?

- (A) 82
- (B) 86
- (C) 88
- (D) 90

Answer: (B) 86

62. What is the value of x if the average of 5 consecutive odd numbers is 90?

- (A) 86
- (B) 90
- (C) 92
- (D) 94

Answer: (B) 90

63. What is the value of x if the average of 5 consecutive odd numbers is 94?

- (A) 90
- (B) 94
- (C) 96
- (D) 98

Answer: (B) 94

64. What is the value of x if the average of 5 consecutive odd numbers is 98?

- (A) 94
- (B) 98
- (C) 100
- (D) 102

Answer: (B) 98

65. What is the value of x if the average of 5 consecutive odd numbers is 102?

- (A) 98
- (B) 102
- (C) 104
- (D) 106

Answer: (B) 102

66. What is the value of x if the average of 5 consecutive odd numbers is 106?

- (A) 102
- (B) 106
- (C) 108
- (D) 110

Answer: (B) 106

67. What is the value of x if the average of 5 consecutive odd numbers is 110?

- (A) 106
- (B) 110
- (C) 112
- (D) 114

Answer: (B) 110

68. What is the value of x if the average of 5 consecutive odd numbers is 114?

- (A) 110
- (B) 114
- (C) 116
- (D) 118

Answer: (B) 114

69. What is the value of x if the average of 5 consecutive odd numbers is 118?

- (A) 114
- (B) 118
- (C) 120
- (D) 122

Answer: (B) 118

70. What is the value of x if the average of 5 consecutive odd numbers is 122?

- (A) 118
- (B) 122
- (C) 124
- (D) 126

Answer: (B) 122

71. What is the value of x if the average of 5 consecutive odd numbers is 126?

- (A) 122
- (B) 126
- (C) 128
- (D) 130

Answer: (B) 126

72. What is the value of x if the average of 5 consecutive odd numbers is 130?

- (A) 126
- (B) 130
- (C) 132
- (D) 134

Answer: (B) 130

Percentages, Averages, and Ratios

Quantitative Aptitude - 3 questions

73. If the price of coal is increased by 25%, by how much percentage must a household reduce its consumption of coal so that the expenditure on coal remains unchanged?
- (A) 15%
(B) 20%
(C) 25%
(D) 33.33%
- Answer: (B) 20%**
74. The average age of a group of 15 miners is 32 years. If the age of the mining supervisor is included, the average age of the group increases by 1 year. What is the age of the mining supervisor?
- (A) 42 years
(B) 45 years
(C) 48 years
(D) 50 years
- Answer: (C) 48 years**
75. In a partnership of coal transportation, A invests $\frac{1}{3}$ of the capital for $\frac{1}{4}$ of the time, B invests $\frac{1}{5}$ of the capital for $\frac{1}{2}$ of the time, and C invests the remaining capital for the whole time. If the total annual profit is Rs. 7,400, what is B's share of the profit?
- (A) Rs. 1,000
(B) Rs. 1,200
(C) Rs. 1,500
(D) Rs. 1,800
- Answer: (A) Rs. 1,000**

Time, Work, Speed, and Distance

Quantitative Aptitude - 10 questions

76. A can complete a piece of work in 12 days, and B can complete the same work in 15 days. They start working together, but A leaves 3 days before the completion of the work. In how many total days is the work completed?
- (A) 6 days
(B) 7 days
(C) 8 days
(D) 9 days
- Answer: (C) 8 days**
77. A train 150 meters long passes a telegraph post in 12 seconds. How long will it take to cross a bridge of length 250 meters at the same speed?
- (A) 20 seconds
(B) 24 seconds
(C) 30 seconds
(D) 32 seconds
- Answer: (D) 32 seconds**
78. Two pipes A and B can fill a tank in 20 minutes and 30 minutes respectively, while a waste pipe C can empty the full tank in 15 minutes. If all three pipes are opened together when the tank is empty, how long will it take to fill the tank to 50% capacity?
- (A) 15 minutes
(B) 30 minutes
(C) 45 minutes
(D) 60 minutes
- Answer: (B) 30 minutes**
79. A man can row upstream at 8 km/h and downstream at 14 km/h. What is the speed of the man in still water, and what is the speed of the stream's current?
- (A) Speed in still water = 11 km/h, Current = 3 km/h
(B) Speed in still water = 10 km/h, Current = 4 km/h
(C) Speed in still water = 12 km/h, Current = 2 km/h
(D) Speed in still water = 9 km/h, Current = 5 km/h
- Answer: (A) Speed in still water = 11 km/h, Current = 3 km/h**

80. Excluding stoppages, the speed of a bus is 54 km/h, and including stoppages, it is 45 km/h. For how many minutes does the bus stop per hour?
- (A) 8 minutes
(B) 10 minutes
(C) 12 minutes
(D) 15 minutes
- Answer: (B) 10 minutes**
81. A, B, and C can complete a task in 10, 12, and 15 days respectively. They started the work together, but A left after 2 days of working. B left 3 days before the completion of the work. How many total days did it take to complete the entire work?
- (A) 5 days
(B) 6 days
(C) 7 days
(D) 8 days
- Answer: (C) 7 days**
82. Two trains start at the same time from Aligarh and Delhi and proceed towards each other at 16 km/h and 21 km/h respectively. When they meet, it is found that one train has traveled 60 km more than the other. What is the distance between Aligarh and Delhi?
- (A) 370 km
(B) 400 km
(C) 444 km
(D) 500 km
- Answer: (C) 444 km**
83. A can do a work in 15 days, which B can do in 20 days. They work together for 6 days, and then the remaining work is completed by C in 2 days. If they receive a total wage of Rs. 3,000 for the entire work, what is the wage share of C?
- (A) Rs. 600
(B) Rs. 900
(C) Rs. 1,000
(D) Rs. 1,200
- Answer: (B) Rs. 900**
84. A motorist covers a distance of 300 km. He travels the first half of the distance at 40 km/h and the second half at 60 km/h. What is his average speed for the entire journey?
- (A) 48 km/h
(B) 50 km/h
(C) 52 km/h
(D) 54 km/h
- Answer: (A) 48 km/h**
85. If 12 men or 18 women can build a wall in 14 days, in how many days can 8 men and 16 women build the same wall working together?
- (A) 6 days
(B) 8 days
(C) 9 days
(D) 10 days
- Answer: (C) 9 days**

Profit, Loss, Interest, and Mixtures

Quantitative Aptitude - 10 questions

86. A shopkeeper sells a mining helmet at a profit of 12%. If he had bought it at 10% less price and sold it for Rs. 18 more, he would have gained 30%. What is the original cost price of the helmet?
- (A) Rs. 360
(B) Rs. 400
(C) Rs. 450
(D) Rs. 500
- Answer: (A) Rs. 360**

87. A sum of money invested at simple interest triples itself in 12 years. In how many years will it become five times itself at the same rate of interest?
- (A) 18 years
(B) 20 years
(C) 24 years
(D) 30 years
- Answer: (C) 24 years**
88. A sum of Rs. 10,000 is lent out in two parts: one at 8% p.a. simple interest and the other at 10% p.a. simple interest. If the total annual interest received is Rs. 920, what is the amount lent at 8% p.a.?
- (A) Rs. 4,000
(B) Rs. 4,500
(C) Rs. 5,000
(D) Rs. 6,000
- Answer: (A) Rs. 4,000**
89. In what ratio must a merchant mix coal of variety A costing Rs. 120 per ton with variety B costing Rs. 150 per ton so that the mixture, when sold at Rs. 161 per ton, yields a profit of 15%?
- (A) 1 : 2
(B) 2 : 1
(C) 1 : 3
(D) 3 : 1
- Answer: (A) 1 : 2**
90. The difference between the compound interest (compounded annually) and the simple interest on a certain sum of money for 2 years at 5% p.a. is Rs. 15. What is the principal sum?
- (A) Rs. 5,000
(B) Rs. 6,000
(C) Rs. 7,500
(D) Rs. 8,000
- Answer: (B) Rs. 6,000**
91. A manufacturer sells an item to a distributor at a profit of 10%, who sells it to a retailer at a profit of 20%, who then sells it to the end consumer at a profit of 25%. If the consumer pays Rs. 1,650 for the item, what was the production cost for the manufacturer?
- (A) Rs. 1,000
(B) Rs. 1,100
(C) Rs. 1,200
(D) Rs. 1,250
- Answer: (A) Rs. 1,000**
92. A, B, and C enter into a partnership. A invests some capital at the beginning. B invests double the amount of A after 6 months, and C invests triple the amount of A after 8 months. If the annual profit is Rs. 27,000, what is the profit share of C?
- (A) Rs. 6,000
(B) Rs. 9,000
(C) Rs. 12,000
(D) Rs. 15,000
- Answer: (B) Rs. 9,000**
93. A vessel is filled with liquid containing 3 parts water and 5 parts syrup. What fraction of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?
- (A) $\frac{1}{3}$
(B) $\frac{1}{4}$
(C) $\frac{1}{5}$
(D) $\frac{1}{7}$
- Answer: (C) $\frac{1}{5}$**
94. A dealer marks his goods 30% above the cost price and then allows a discount of 10% on the marked price. What is his net profit percentage?
- (A) 15%
(B) 17%
(C) 20%
(D) 23%
- Answer: (B) 17%**

95. What is the compound interest on a sum of Rs. 25,000 for 1.5 years at 12% p.a. compounded half-yearly?
- (A) Rs. 4,620
(B) Rs. 4,775.40
(C) Rs. 4,800
(D) Rs. 4,912.20
- Answer: (B) Rs. 4,775.40**

Syllogisms and Venn Diagrams

Reasoning Ability - 2 questions

96. Statements:
I. Some coal mines are deep.
II. All deep mines are dangerous.
- Conclusions:
1. Some dangerous mines are coal mines.
2. No deep mine is a coal mine.
- Which of the conclusions logically follows?
- (A) Only conclusion 1 follows
(B) Only conclusion 2 follows
(C) Both conclusions 1 and 2 follow
(D) Neither conclusion 1 nor 2 follows
- Answer: (A) Only conclusion 1 follows**

97. Choose the Venn diagram that best represents the relationship among the following classes:
Coal, Carbon, Diamond
- (A) Three disjoint circles
(B) One circle inside another, which is inside a third circle
(C) Two concentric circles inside a larger third circle, with the two inner circles being completely disjoint
(D) Two intersecting circles inside a larger third circle
- Answer: (C) Two concentric circles inside a larger third circle, with the two inner circles being completely disjoint**

Coding, Directions, and Relations

Reasoning Ability - 10 questions

98. If in a certain language, 'COMPILER' is coded as 'OCPMILRE', how is 'DATABASE' coded in that same language?
- (A) ADTABASE
(B) ADATABES
(C) ADTABAES
(D) ADATBAES
- Answer: (B) ADATABES**
99. A person starts from office, walks 10 km towards South, then turns left and walks 15 km, then turns left and walks 18 km, and finally turns left and walks 15 km. How far and in which direction is he from his starting point?
- (A) 8 km, South
(B) 8 km, North
(C) 10 km, North
(D) 12 km, South
- Answer: (B) 8 km, North**
100. Introducing a boy, a girl said, 'He is the son of the daughter of the father of my uncle.' How is the boy related to the girl?
- (A) Brother
(B) Nephew
(C) Uncle
(D) Son-in-law
- Answer: (A) Brother**

101 If 'A + B' means A is the brother of B; 'A - B' means A is the sister of B; and 'A * B' means A is the father of B. Which of the following equations represents that 'C is the nephew of D'?

- (A) $D - E + C * F$
- (B) $D + E * C - F$
- (C) $D - E + C - F$
- (D) $D + E * C + F$

Answer: (D) $D + E * C + F$

102 Find the missing number in the following series:

4, 9, 20, 43, 90, ?

- (A) 180
- (B) 183
- (C) 185
- (D) 187

Answer: (C) 185

103 A rat runs 20 m towards East and turns to right, runs 10 m and turns to right, runs 9 m and again turns to left, runs 5 m and then turns to left, runs 12 m and finally turns to left and runs 6 m. Which direction is the rat facing now?

- (A) East
- (B) West
- (C) North
- (D) South

Answer: (C) North

104 In a certain code language, 'pit na som' means 'bring me water', 'na jo tod' means 'water is life', and 'tub od pit' means 'bring some food'. Which word in that language represents 'me'?

- (A) pit
- (B) na
- (C) som
- (D) jo

Answer: (C) som

105 Point Q is 12m South of Point P. Point R is 8m West of Point Q. Point S is 5m North of Point R. Point T is 8m East of Point S. How far and in which direction is Point T with respect to Point P?

- (A) 7m, South
- (B) 7m, North
- (C) 5m, South
- (D) 12m, West

Answer: (A) 7m, South

106 A is the father of B. C is the sister of B. D is the mother of A. E is the sister of D. How is C related to D?

- (A) Grandmother
- (B) Granddaughter
- (C) Mother
- (D) Aunt

Answer: (B) Granddaughter

107 If 'REQUEST' is written as 'S2R52TU', how is 'ACID' written in that same code?

- (A) 1D3E
- (B) BDJE
- (C) 13JD
- (D) 1D3D

Answer: (A) 1D3E

Arrangements, Puzzles, and Deductions

Reasoning Ability - 10 questions

108 Six friends A, B, C, D, E, and F are sitting in a circle facing the center. B is between F and D; E is between A and C. If A is to the immediate left of F, who is sitting opposite to B? (Assume uniform circular spacing).

- (A) A
- (B) C
- (C) D
- (D) E

Answer: (D) E

109 Four girls A, B, C, and D are sitting on a bench facing North. A is to the immediate left of B. C is between B and D. Who is sitting on the extreme left end of the bench?

- (A) A
- (B) B
- (C) C
- (D) D

Answer: (A) A

110 Statements:

- I. The local municipal corporation has announced a complete ban on plastic bags below 50 microns.
- II. Local citizens are actively organizing cleanliness drives and collecting plastic wastes from water bodies.

Choose the correct relation between the statements:

- (A) Statement I is the cause and statement II is its effect.
- (B) Statement II is the cause and statement I is its effect.
- (C) Both statements I and II are independent causes.
- (D) Both statements I and II are effects of some common cause.

Answer: (D) Both statements I and II are effects of some common cause.

111 Statement: 'A double-blind clinical trial is the gold standard for testing the safety and efficacy of any new vaccine before mass public rollout.'

Assumptions:

- 1. Vaccines cannot be rolled out to the public without testing.
- 2. Double-blind trials are a reliable methodology to evaluate vaccines.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption 1 is implicit
- (B) Only assumption 2 is implicit
- (C) Both assumptions 1 and 2 are implicit
- (D) Neither assumption 1 nor 2 is implicit

Answer: (C) Both assumptions 1 and 2 are implicit

112 Five people P, Q, R, S, and T are standing in a queue. R is taller than S but shorter than P. Q is the tallest. T is slightly shorter than R but slightly taller than S. Who is the second shortest person in the group?

- (A) P
- (B) R
- (C) S
- (D) T

Answer: (D) T

113 Statement: 'Should the government completely stop subsidizing fossil fuels and divert those funds to renewable energy research?'

Arguments:

- 1. Yes, this will accelerate the transition to green energy and significantly lower carbon emissions.
- 2. No, a sudden complete removal of subsidies will cause a massive spike in transportation costs and fuel prices, causing severe inflation and hurting the poorest citizens.

Which of the arguments is/are strong?

- (A) Only argument 1 is strong
- (B) Only argument 2 is strong
- (C) Both arguments 1 and 2 are strong
- (D) Neither argument 1 nor 2 is strong

Answer: (C) Both arguments 1 and 2 are strong

114 In a coding system, if 'BLUE' is coded as 40 and 'RED' is coded as 27, what is the code value for 'GREEN'?

- (A) 44
- (B) 49
- (C) 51
- (D) 53

Answer: (B) 49

115Statement: 'A heavy landslide near a coal mine has completely blocked the main access highway, trapping several logistics trucks.'

Courses of Action:

1. The district administration should immediately dispatch heavy earth-moving machinery to clear the debris and restore traffic.
2. All mining operations should be permanently shut down to prevent any future landslide risks.

Which of the courses of action should be followed?

- (A) Only course 1 should be followed
- (B) Only course 2 should be followed
- (C) Both courses 1 and 2 should be followed
- (D) Neither course 1 nor 2 should be followed

Answer: (A) Only course 1 should be followed

116Six books P, Q, R, S, T, and U are placed side-by-side on a shelf. Q is between U and P. S is between P and T. R is at the extreme right end. U is to the immediate right of R? (Wait, let's re-verify: if R is at the extreme right, U cannot be to the right of R. Let's design a consistent 6-book arrangement):

Let's place the books at slots 1, 2, 3, 4, 5, 6 from left to right:

- R is at the extreme left end (slot 1).
- U is to the immediate right of R, so U is at slot 2.
- Q is between U and P, so Q is at slot 3, and P is at slot 4.
- S is between P and T, so S is at slot 5, and T is at slot 6.

This arrangement is perfectly linear: [R, U, Q, P, S, T]!

Let's check the clues:

- R is at the extreme left end (slot 1).
- U is to the immediate right of R (slot 2).
- Q is between U and P (slot 3, between 2 and 4).
- S is between P and T (slot 5, between 4 and 6).

Which book is at the extreme right end?

- (A) P
- (B) S
- (C) T
- (D) U

Answer: (C) T

117Is the integer 'x' an odd number?

Statements:

- I. x is a multiple of 3.
- II. x is the sum of two consecutive prime numbers.

Choose the correct data sufficiency parameter:

- (A) Statement I alone is sufficient, but Statement II alone is not.
- (B) Statement II alone is sufficient, but Statement I alone is not.
- (C) Both Statements I and II together are sufficient, but neither alone is.
- (D) Statements I and II together are NOT sufficient, and more data is needed.

Answer: (C) Both Statements I and II together are sufficient, but neither alone is.

